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RESEARCH ARTICLE

MEDICAL PHYSICS

Development and multi-institutional validation of a convolutional neural network to detect vertebral body mis-alignments in 2D x-ray setup images

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Abstract

Background: Misalignment to the incorrect vertebral body remains a rare but serious patient safety risk in image-guided radiotherapy (IGRT).
Purpose: Our group has proposed that an automated image-review algorithm be inserted into the IGRT process as an interlock to detect off-by-one vertebral body errors. This study presents the development and multi-institutional validation of a convolutional neural network (CNN)-based approach for such an algorithm using patient image data from a planar stereoscopic x-ray IGRT system.
Methods: X-rays and digitally reconstructed radiographs (DRRs) were collected from 429 spine radiotherapy patients (1592 treatment fractions) treated at six institutions using a stereoscopic x-ray image guidance system. Clinically-applied, physician approved, alignments were used for true-negative, “no-error” cases. “Off-by-one vertebral body” errors were simulated by translating DRRs along the spinal column using a semi-automated method. A leave-one-institution-out approach was used to estimate model accuracy on data from unseen institutions as follows: All of the images from five of the institutions were used to train a CNN model from scratch using a fixed network architecture and hyper-parameters. The size of this training set ranged from 5700 to 9372 images, depending on exactly which five institutions were contributing data. The training set was randomized and split using a 75/25 split into the final training/validation sets. X-ray/ DRR image pairs and the associated binary labels of “no-error” or “shift” were used as the model input. Model accuracy was evaluated using images from the sixth institution, which were left out of the training phase entirely. This test set ranged from 180 to 3852 images, again depending on which institution had been left out of the training phase. The trained model was used to classify the images from the test set as either “no-error” or “shifted”, and the model predictions were compared to the ground truth labels to assess the model accuracy. This process was repeated until each institution’s images had been used as the testing dataset.
Results: When the six models were used to classify unseen image pairs from the institution left out during training, the resulting receiver operating characteristic area under the curve values ranged from 0.976 to 0.998. With the specificity fixed at 99%, the corresponding sensitivities ranged from 61.9% to

99.2% (mean: 77.6%). With the specificity fixed at 95%, sensitivities ranged from 85.5% to 99.8% (mean: 92.9%).
Conclusion: This study demonstrated the CNN-based vertebral body misalignment model is robust when applied to previously unseen test data from an outside institution, indicating that this proposed additional safeguard against misalignment is feasible.

KEYWORDS
convolutional neural networks, patient safety, planar x-ray setup images

1 INTRODUCTION

The field of radiotherapy is remarkably safe, with crude error rates per delivery fraction of 0.06%–0.29% reported in the literature. However, misalignment of radiation treatments to the patient anatomy, even when using image-guided radiotherapy (IGRT), remains an important source of error. The Radiation Oncology Incident Learning System (RO-ILS) was established in 2014 as an inter-institutional error reporting system. In a 2018 report, 396 incidents were investigated, of which 40 were classified as "wrong shift instructions given to therapists," and 34 as "wrong shift performed at treatment.[1]" The Q3 2016 RO-ILS summary report describes a case of a treatment misaligned by 3 cm and opined: "This is one of approximately 28 IGRT events documented this quarter. Clearly we need to increase the attention paid to how to decrease the number of IGRT-related issues throughout the field.[2]" Off-by-one vertebral body errors are particularly insidious due to the translational symmetry of the vertebral column.[3] Redundant safety checks are in place during the simulation imaging, treatment planning, setup imaging, and treatment delivery phases, with the overall goal of catching potential errors and correcting them before treatment is actually delivered to the patient. Identification of error pathways,[4–6] the use of established safety protocols,[7,8] and redundant safety checks[9] are common approaches to radiation therapy error mitigation. However, ample evidence demonstrates inadequate compliance with safety protocols in practical healthcare situations. Recent studies have reported surgical safety checklist compliance of 52–80%.[10–12] In a 2017 report of adverse events in Minnesota,[13] the most commonly identified root cause category was "Rules/Policies/Procedures", and within that category the most commonly reported factor was "policies/procedures are in place, but not followed". Mallet et al.[14] studied eight cases of wrong site/procedure/patient events occurring from 2008–2010 at an academic medical center. They stated "the Rules, Policies, and Procedures category contained the highest number of failure modes (22) and was present in all eight wrong-person, procedure, and site events analyzed. *Every failure mode within this category reflects an incomplete or improper use of the 3 steps involved with the Universal Protocol* (emphasis added)."

In this work we focus on what has been termed an "IGRT Error", in which human error contributes to misalignment of a radiotherapy treatment even with the use of image guidance. Setup imaging prior to each fraction of radiation is an important part of the safety protocols in place to ensure that the patient is correctly aligned and that the radiation is delivered to the appropriate target. However, IGRT still depends on manual verification of patient alignment, a process that can be improved upon with the introduction of a quantitative image alignment metric.[15] One study found that patient setup errors accounted for almost half of all documented incidents at their institution over a 10 year period.[16] Many different imaging modalities can be used to perform setup imaging, but in this work we focus on the planar x-ray setup imaging system ExacTrac.[17] This type of imaging is commonly used for patients receiving high dose radiation to the cranial or spinal regions due to its high precision and submillimeter alignment accuracy.[18] For ExacTrac imaging, two stereoscopic x-ray images of the patient are acquired after initial positioning on the treatment couch. These x-rays are then matched to corresponding digitally reconstructed radiographs (DRRs) that are generated using the simulation CT image to determine what translational and rotational shifts need to be done to the patient to align them properly before treatment. The ExacTrac system allows for high precision alignment on the order of submillimeter accuracy[17]; however, due to a relatively small field of view and the possibility of an image similarity-based minimization process to lock on to local minima in the cost function,[19] the system can align to an adjacent vertebral body instead of the one intended for treatment. This effect is particularly pronounced in the lower cervical spine and thoracic spine since adjacent vertebral bodies in these regions can be difficult to visually differentiate.[20,21] A wrong level vertebral body mismatch can have dire consequences for the patient undergoing treatment including reduced tumor control as well as increased normal tissue damage. Some institutions attempt to mitigate this risk factor by pairing ExacTrac imaging with additional imaging to provide a redundant check on the patient's treatment position, but even with this safety measure in place errors can and do still occur.

Due to the limitations of protocol- and redundant check-based IGRT error mitigation, our group proposed that an automated image-review algorithm could be inserted into the IGRT process to act as an interlock to detect and prevent IGRT errors.[22] While deep learning algorithms have been introduced to many parts of the radiation oncology patient treatment and quality assurance workflows,[23–26] to our knowledge our work is the first to tackle the application of deep learning as a failsafe to strengthen the human effort of IGRT image review. Previous work applying deep learning to computer vision problems in the spine has focused on approaches for automatic segmentation,[27,28] detection,[29] diagnosis,[30,31] and even motion monitoring,[32] with good results. Furthermore, since the mid-2010s convolutional neural networks have been shown to greatly outperform other methods in image classification tasks,[33] leading us to choose this methodology for our own work. Our work develops the novel application of a deep learning-based tool for the automatic detection of patient misalignments in IGRT. Here we introduce a highly accurate deep learning-based approach to detect off-by-one vertebral body misalignments developed and validated using a multi-institutional patient dataset. By assembling a dataset consisting of images from a multi-institutional collaboration, our work overcomes a common limitation of artificial intelligence algorithms, namely the lack of robustness due in large part to a lack of data.[34],[35] This problem was called out specifically by Qu et al. who argue that "shared huge datasets are needed" in order to overcome the current limitations of deep learning-based spine image analysis.[36] Our goal in this work is to present a robust and validated deep learning-based solution to the known problem of vertebral body misalignments in the ExacTrac imaging setup system.

2 METHODS AND MATERIALS

2.1 Data collection and curation

Anonymized patient data was collected from all six participating institutions under the auspices of an approved Institutional Review Board (IRB) research protocol at the coordinating institution (Institution 1). Each institution searched their own internal database to identify thoracic spine patients treated using the ExacTrac image guidance system. In all, 429 patient datasets were collected. Table 1 details the datasets provided by each institution.

For each of these patients, ExacTrac x-ray/DRR pairs from each treatment fraction were anonymized and collected into a central image database at the coordinating institution. If multiple x-ray/DRR pairs were present in a single fraction (typically the case), the final x-ray/DRR pair from that treatment fraction was used. The final database consisted of 3184 x-ray images, representing 1592 individual treatment fractions since each fraction generates two x-ray setup images, as well as the corresponding DRRs. A representative x-ray/DRR pair is shown in Figure 1. It was visually validated that these clinically-approved image sets did not inadvertently contain misaligned images (i.e. actual treatment errors).

Simulated off-by-one vertebral body misalignments were then created for each x-ray/DRR pair using a semi-automated method based on a grid search of local maximum values of the cross-correlation coefficient. The x-ray image and DRR were each first down-sampled by a factor of four. The down-sampled DRR was then shifted pixel-wise against the stationary x-ray image, and the cross-correlation coefficient computed for each of these shifts. Cross-correlation coefficients were used as the image similarity metric since they exhibit good performance in the 2D to 3D

TABLE 1 Dataset statistics for the multi-institutional collaboration.

Institution	# of Patients	# of Fractions	# of clinical images	Date range
Institution 1	87	330	660	6/2014 – 12/2017
Institution 2	72	246	492	10/2017 – 10/2021
Institution 3	45	228	456	4/2015 – 7/2021
Institution 4	38	116	232	7/2020 – 11/2021
Institution 5	174	642	1,284	7/2020 – 11/2021
Institution 6	13	30	60	12/2013 – 1/2021

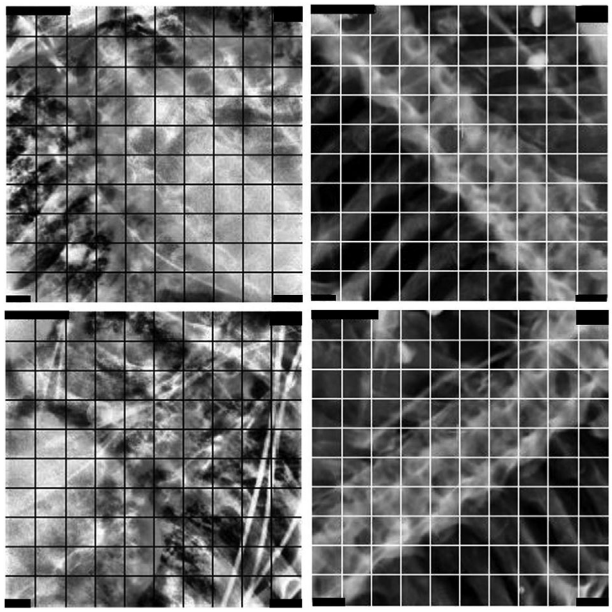

FIGURE 1 X-rays (left) and DRRs (right) from an example properly aligned thoracic spine patient.

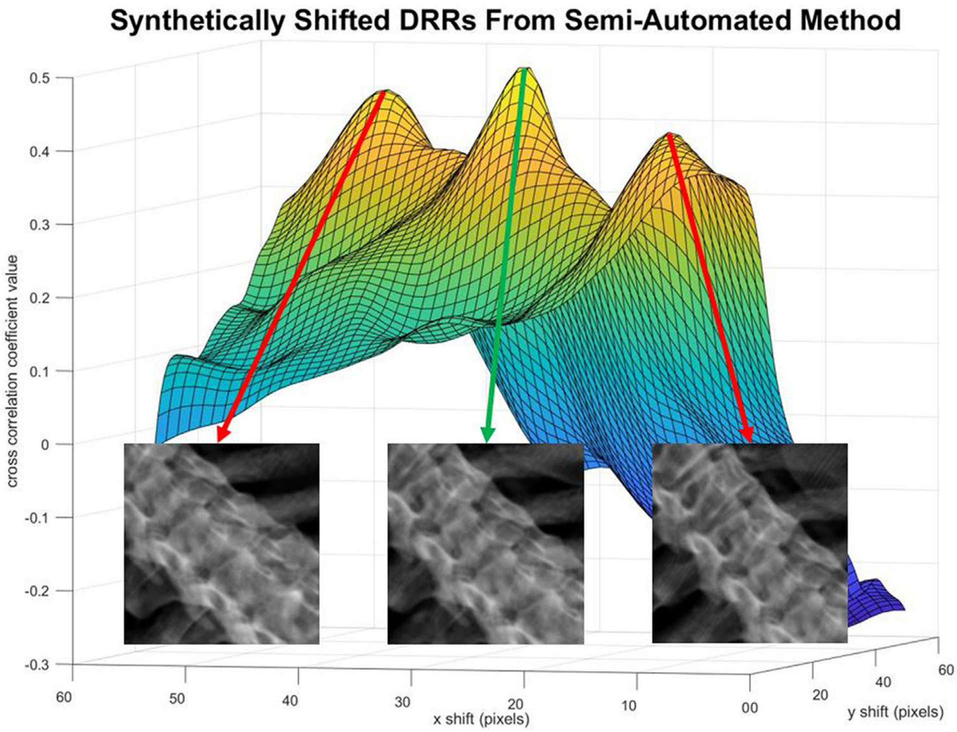

FIGURE 2 Synthetic shifted DRRs (red arrows) obtained by shifting the correctly aligned DRR (green arrow) in both directions along the spinal column and realigning to a local maximum of the cross-correlation coefficient as shown by the surface plot of cross-correlation coefficient values.

medical image registration domain.[37] Finally, the local maximum values were labeled on the grid view of all cross-correlation coefficients. From this map of values, the two local maximum points representing a shift in each direction along the spinal column were manually selected. These shifted DRRs were verified against the original x-ray image to ensure that the images did indeed represent a visual shift by one vertebral body. Figure 2 illustrates the semi-automated method and shows the resulting shifted DRRs that were generated for a representative patient case. In some instances, such as patients with spine fixation hardware present or a targeted vertebral body with a visible pathology in the images, the position of the local maximum did not correspond to an off-by-one vertebral body shift as judged by visual interpretation. For those images, the vertebral body shifts were performed manually without the aid of the cross-correlation coefficient values. While these cases likely represent a relatively easy target for the error detection algorithm, we believed it was appropriate to include the entire data sample as representative of the population of treated patients. A graphical user interface (GUI) was used to blend between the shifted DRR and unshifted x-ray to ensure that the chosen shift was visually reasonable. All images were cropped to maintain uniform image dimensions across the two classes of shifted and non-shifted images. Of note, each of the two stereoscopic images from the ExacTrac image guidance system was treated independently for this study. Finally, two-channel image arrays were created out of the final set of shifted and non-shifted images. The final set of DRRs contained one non-shifted, one shifted up by one vertebral body, and one shifted down by one vertebral body, for each original image pair. Each of these three DRRs was matched with the corresponding (non-shifted) x-ray. The final dataset consisted of 9,552 such 2-channel image arrays spanning the six collaborating institutions.

2.2 | Data organization

Once the x-ray/DRR pairs were collected from each institution and synthetic alignment errors generated, the final dataset of 9552 two-channel image arrays was organized in two distinct ways. Our primary objective was to analyze the robustness of our model using a "leave one out" approach, where the model was trained using data from all institutions but one and tested on data from the institution left out. Our secondary objective was to compare the classification accuracy from a pooled multi-institutional model to that of a single-institutional model, where for each model the training and test datasets were organized in such a way so as to ensure that no patient from any institution ended up in both datasets.

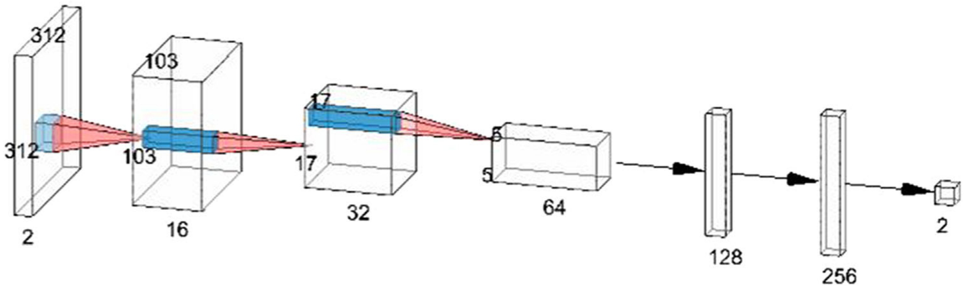

FIGURE 3 Purpose-built CNN architecture for x-ray/ DRR vertebral body image pair classification.

For the “leave one out” approach, all of the two-channel images from five of the six collaborating institutions were collected into a single training dataset. The size of the dataset ranged from 5700 to 9372 distinct two-channel image arrays depending on which five of the institutions were used. This training dataset was then randomized and split into training and validation datasets using a 75/25 training/validation split. The testing dataset consisted of the images from the sixth institution, which had been left out of the training phase.

For the pooled multi-institutional model, two steps were taken to create the image datasets. First, the images from each of the six individual institutions were divided into training and test datasets using an approximately 80/20 split. The split was performed on the data organized in alphabetical order by anonymized name label, with no randomization. When the exact 80/20 split occurred in the middle of a set of images from the same patient, all images from that patient were grouped into the training dataset. This was done to ensure that patients who appeared in the training dataset would not appear in the test dataset (even if they were treated with multiple fractions) because this would potentially bias the evaluation of the final model accuracy. Second, the training datasets from all institutions were compiled into a training dataset of 7686 distinct two-channel image arrays, then randomized and split into final training/validation datasets using a 75/25 split. The training dataset from Institution 1 (the coordinating institution), consisting of 1626 image arrays, was duplicated and saved separately prior to compiling for use in the single-institution model. The testing dataset from Institution 1 alone consisted of 354 image arrays while the pooled testing dataset from all six institutions consisted of 1866 image arrays.

2.3 Model design

We arrived at our final model architecture (shown in Figure 3) after first exploring multiple machine learning methods for our specific classification problem. We used only data from Institution 1 to first investigate the most accurate model to use before moving to the full multi-institutional dataset. Three models were developed: a logistic regression model, a pre-trained CNN that was adapted to our application using transfer learning, and a CNN trained from scratch. Details on all three models can be found in Supplementary Document 1. All three models were trained using the same spine image training dataset from Institution 1. When the purpose-built CNN for Institution 1’s data was used to classify the previously unseen test image pairs, the resulting AUC was 0.972. For comparison, the transfer learning-based CNN and the logistic regression models tested on the same single-institution test image dataset obtained AUCs of 0.876 and 0.801, respectively. With the specificity fixed at 99%, the purpose-built CNN achieved a sensitivity of 64.5% in correctly classifying shifts of one vertebral body as compared to a sensitivity of 32.3% for the transfer learning-based CNN and 23.7% for the logistic regression model. From these results, it was determined that a neural network trained from scratch would give the highest classification accuracy for our particular problem of detecting vertebral body misalignments.

Model design and hyper-parameter tuning were performed on data from Institution 1. Network architectures of various depths were investigated and optimal classification results were achieved with a five convolutional block neural network similar to that of AlexNet.[33] While increasing CNN depth has been shown to improve final classification accuracy[38] when trained on datasets with millions of images, it also increases the possibility of overfitting when used with smaller datasets such as ours. Convolutional layers were followed by rectified linear activation functions and batch normalization layers. Max pooling layers were interspersed with convolutional layers. Two dense layers, each of which was followed by a 50% dropout layer to reduce overfitting,[39] were used before the final classification layer. The learning rate was set to 1e-4.[40] The Adam optimizer[41] and sparse categorical cross-entropy loss function were used for training. These hyper-parameters were tuned using Institution 1’s data and remained unchanged for all subsequent multi-institutional model training. Early stopping was implemented on all models, again with the aim

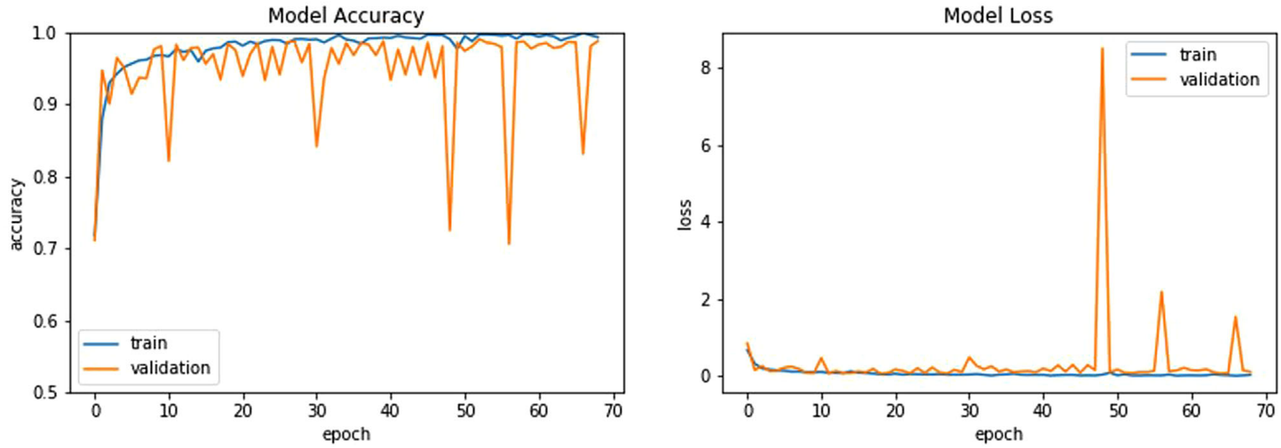

FIGURE 4 Model accuracy (left) and loss (right) as a function of training epoch for a representative model training with Institution 6 left out of the training and validation sets. Early stopping was implemented and model training stopped after 69 epochs for this particular training iteration, with the model from epoch 19 being saved and used for further analysis.

of reducing model overfitting. The training and validation accuracy and loss as a function of training epoch are shown in Figure 4 for an example model training.

2.4 | Model training

We designed and trained a series of convolutional neural networks (CNNs) to perform the binary classification task of discriminating between shifted and non-shifted images. These CNNs took as input the two-channel arrays described in the Data Collection and Curation section above, where each array was labeled as either "Shifted" or "Unshifted" and consisted of either correctly aligned or mis-aligned x-ray/DRR pairs. After the CNN had been trained, it was tested on the reserved two-channel arrays. These arrays were input into the CNN, and the model would output a score from 0 to 1 based on how likely it thought the images were to represent a patient misalignment.

Once the CNN architecture with the optimal performance on Institution 1's data had been identified and training hyper-parameters set, the CNN was retrained on the multi-institutional image data in two different ways:

First, we trained six models using a "leave one institution out" approach to estimate the accuracy of the model when applied to an entirely new institution's data during the testing phase. In this step, we re-trained a model using the CNN architecture and parameters described in the following section, with the training data consisting of all images from five of the six different institutions as described above. Early stopping was implemented in all of our models with the aim of reducing overfitting, but all other hyper-parameters remained unchanged once they had been optimized using Institution 1's data. At testing time, the trained model was applied to all images from the institution that had been left out of the training dataset to evaluate robustness to an outside institution's image data. This process was repeated to generate six models, each one representing each institution being left out of the training dataset.

To establish the necessity of using multi-institutional data for this error detection algorithm, we compared the CNN's performance when trained on multi-institutional versus single-institutional data. The CNN was trained using the training dataset of 1626 x-ray/DRR pairs from Institution 1. A pooled model was created by training the CNN on the pooled training dataset composed of 7686 image arrays from all six institutions. The single-institution model was tested separately on the reserved set of patient data from Institution 1, and on the pooled test set described above. The pooled model was tested on the pooled test set.

2.5 | Data analysis

For both the pooled model and the six "leave one institution out" models, receiver operating characteristic (ROC) curves were used to evaluate model performance. From these curves, the area under the curve (AUC) is reported as a measure of overall model accuracy. The network outputs a continuous variable in the range of 0 to 1 representing a likelihood of image misalignment. Clinical implementation as a binary error-detection classifier would require application of a threshold, which would depend on what was deemed an acceptable sensitivity-specificity tradeoff. Sensitivity results for three different specificity values of interest, namely, 99%, 95%, and 90%, are also reported. A specificity of 99% would correlate to approximately one false positive per treatment machine per week under a typical treatment load, assuming an automated error detection algorithm was applied to every treatment—an acceptable rate of false positive disruptions compared to other safety interlocks in common use. We focus on this high specificity threshold due to the well-documented patient safety implications of alert fatigue in healthcare.[42,43] While alarm fatigue is not the only factor that can contribute to decreased staff performance in a healthcare setting,[44] it is nonetheless still recognized as an important factor. Recent work has highlighted the lack of research into alarm fatigue within radiation oncology specifically,[45] and attempted to raise awareness of this problem. Specificity of 95% or 90%, corresponding to 5% and 10% false positive rates, could be appropriate in the context of stereotactic spine radiotherapy, which is a relatively infrequent and high-risk procedure. The choice of an appropriate threshold at which an alarm is triggered is an important step in reducing alarm fatigue in the clinic.[43]

We also analyzed the false positive rate when small shifts, on the order of a few pixels, were present in the test data. We used the final leave-one-out model where Institution 1 was left out of the training dataset. We first tested the final model on all images from Institution 1 to obtain the ROC curves and the sensitivity values at a few set specificity values, as described just above. We obtained the threshold value corresponding to the 95% specificity level for this analysis. Twelve patients, representing 31 individual treatment fractions, were randomly selected from Institution 1. For each fraction, both horizontal and vertical shifts of ±1 pixel, ±2 pixels, ±3 pixels, ±5 pixels, and ±8 pixels were then applied to each DRR. For the ExacTrac system, five pixels corresponds to a shift of approximately 1 mm, while eight pixels corresponds to a shift of approximately 1.5 mm. Finally, the trained model was applied to all of the shifted images, and any prediction value above the threshold set by the 95% specificity level was labeled as a false positive.

3 RESULTS

When the six "leave one institution out" models were used to classify image pairs from the institution left out during training, the resulting AUC values ranged from 0.976 to 0.998 (Figure 5). With the specificity fixed at 99%, the corresponding sensitivities ranged from 61.9% to 99.2% with a mean of 77.6% (Table 2). When the specificity was set at 95%, corresponding sensitivities from 85.5% to 99.8% (mean: 92.9%) were observed. The median threshold value required for the 95% specificity set point was 0.682.

When the model trained on data from Institutions 2–6 was applied to the 12 patients from Institution 1 with clinically insignificant misalignments, the false positive rate did not differ appreciatively from that seen when it was applied to the clinical (no error) images. For these 12 patients, the false positive rate was 3.2% on the unshifted images, 4.0% on the images shifted

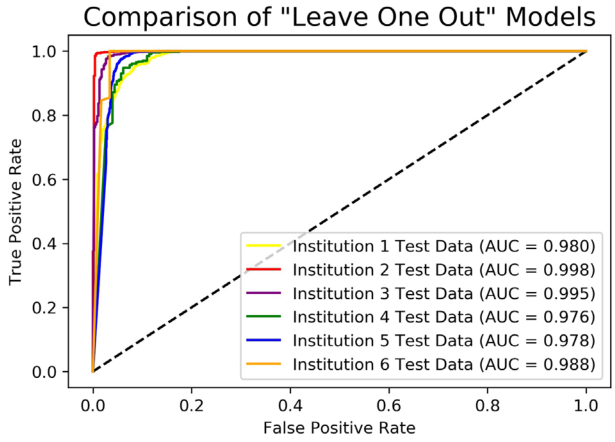

FIGURE 5 Comparison of classification performance in correctly identifying shifts of one vertebral body among the six distinct "leave one institution out" models. Each label describes the institution that was left out of training and used as the test dataset.

TABLE 2 Corresponding sensitivities for the neural network output value thresholds corresponding to 90%, 95%, and 99% specificity, respectively, for the six "leave one institution out" models.

Institution used For testing	% Sensitivity at 90% specificity	% Sensitivity at 95% specificity	% Sensitivity at 99% specificity
Institution 1	95.8	87.5	61.9
Institution 2	99.9	99.8	99.2
Institution 3	99.8	98.9	79.8
Institution 4	96.7	89.6	75.0
Institution 5	99.8	96.0	65.2
Institution 6	100.0	85.5	84.6

by ± 1 pixel, 5.6% on the images shifted by ± 2 pixels, and 6.9% on the images shifted by ± 3 pixels. The model flagged 19.4% of the images shifted by 5 pixels (approximately 1 mm) and 51.2% of the images shifted by 8 pixels (approximately 1.5 mm) as misaligned. It is worth noting that in many institutions, a shift of 1 mm is considered to be the upper end of the typical tolerance used for these high-dose stereotactic spine regimens. These results are shown in Figure S1.

When the purpose-built CNN was trained and tested using only Institution 1's data, it obtained an AUC of 0.975 (Figure S2). When this model trained on a single institution's data was applied to a multi-institutional test set, the AUC dropped to 0.942. By comparison, a model with the same network architecture and hyperparameters, but trained and tested using pooled data from all collaborators, obtained an AUC of 0.992. Sensitivity and specificity threshold values for all three of these models are reported in Table 3. It is worth noting that at the fixed specificity of 99%, the model trained using only a single institution's image data obtained a sensitivity of just 67.9% (single-institution test set) or 57.4% (multi-institution test set) as compared to

TABLE 3 Corresponding sensitivities for the neural network output value thresholds corresponding to 90%, 95%, and 99% specificity for the three models trained and tested using the datasets shown in the leftmost column. The complete ROC curves are shown in Figure S2.

Training dataset/Testing dataset	% Sensitivity at 90% specificity	% Sensitivity at 95% specificity	% Sensitivity at 99% specificity
Single-institution/ Single-institution	95.3	91.9	67.9
Single-institution/ Multi-institution	85.6	77.4	57.4
Multi-institution/ Multi-institution	99.3	97.7	79.3

79.3% for the multi-institutional model. When the specificity was fixed at 95%, the single-institution model obtained sensitivities of 91.9% and 77.4% on the single- and multi-institutional test sets, respectively, and the multi-institution model obtained a sensitivity of 97.7%.

4 | DISCUSSION

Based on the results presented above, application of this error detection model to data from an unseen institution would result in an error detection sensitivity of at least 86%–100% (mean: 93%) if a false positive rate of approximately 5% was accepted, or a sensitivity of 62%–99% (mean: 78%) if only a 1% false positive rate was accepted. We believe these results demonstrate sufficient accuracy to warrant clinical implementation of this error detection model. The corresponding effort to deal with the level of false positives flagged by this model is small compared to other safety processes already in use and would be expected to reduce IGRT errors by approximately a factor of 5. If up to 5% false positive rate was allowed, which could be acceptable given the relative infrequency of spinal radiotherapy (particularly high-risk stereotactic regimens), then errors could be reduced by a factor as large as 10.

However, further improvements in accuracy would allow this error detection algorithm to truly run in the background and only disturb the workflow in case of an actual error. Because relatively few thoracic spine patients are treated with ExacTrac at any one of our institutions in a given year, the size of our training dataset was relatively small in comparison to the truly big data such as ImageNet,[46] a database containing over 10 million images which is used to train state-of-the-art image classification models. Further increasing the amount of training data, as well as the number of institutions involved, could be expected to improve upon the accuracy and robustness of our error detection model. Others have proposed the use of transfer learning to adapt models trained on ImageNet to the medical domain where high-quality training data is more limited.[47] We found in our specific implementation that transfer learning was not as effective as training from scratch, but this avenue warrants further exploration. Finally, while the stereoscopic imaging system generates a set of two oblique x-ray images, for the purposes of this work we considered each of these images independently. This limitation arose due to challenges in simulating vertebral body misalignments that preserve geometric correlations between the two stereoscopic images. Future work will focus on methods for treating the two x-ray images and the associated DRRs as a single, interdependent image set, potentially increasing the error-detecting power of the model.

We incorporated multi-institutional data into the development of our error-detection model because we anticipated that it would lead to better accuracy on data from unseen institutions. This expectation was validated by the results of our leave-one-institution-out methodology. We can infer the presence of qualitative differences in ExacTrac image data from participating institutions by differences in algorithm sensitivity of as much as 14% at a 95% specificity threshold. All institutions utilize similar default imaging parameters for their thoracic spine patients (25 mAs and 120 or 130 kVp). A trend towards higher sensitivity for institutions with the higher default kVp value was observed, but no statistically significant correlations were observed. We were not able to identify obvious visual features in the data samples that might explain those differences. This is highlighted in Figure S3 showing some representative false positive images flagged by the different leave-one-institution-out models at the 95% specificity threshold for each model. We computed the distributions of the average x-ray pixel values for each institution, and found that the distribution for Institution 2 differed significantly from the other institutions (p-value < 0.05) when a Mann-Whitney U test was performed. The leave-one-out model where Institution 2 was left out of the training dataset and used as the test set achieved the highest performance, and the statistically significant difference in average x-ray pixel values suggests a real difference in the images. However, the exact nature and underlying cause of that difference is unknown and beyond the scope of the current work. Possibilities include imager performance, differences in patient populations, or a different mix of spine levels treated at each institution. At this time, all participating institutions are academic and/or high-volume medical centers that are considered expert institutions in spine radiotherapy. These data may not be representative of ExacTrac image data at all clinics in the radiotherapy community, and the model performance has not been tested on data from institutions that may image using suboptimal imaging techniques (improper mAs or kVp, for example).

The power of automation to promote radiotherapy safety and efficiency is widely recognized, but most research to date has focused on automation in machine quality assurance,[48] treatment planning[49] and plan evaluation,[50] and auto-contouring.[51] However, little has been done to incorporate automation and machine learning into the task of image review, which is a function that requires a significant amount of effort from radiation oncologists, physicists and technologists in the IGRT workflow.[52,53] We believe our work adds significantly to this effort. Future work should focus on using implementation science to optimize the integration of such a tool into the current radiation oncology clinical workflow, since currently unstudied barriers to practical use likely exist.[54]

5 | CONCLUSIONS

In this work we have developed a convolutional neural network (CNN)-based model for the automatic detection of vertebral body misalignments in planar x-ray setup images. We believe such an algorithm could be integrated into the treatment workflow, either directly within an image-guided radiotherapy system or as a standalone ancillary system, to provide a software interlock to prevent treatment of the incorrect vertebral body. Our results demonstrated that this misalignment detection model is robust when applied to previously unseen test data from an outside institution, indicating that this proposed additional safeguard against misalignment is feasible.

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CONFLICT OF INTEREST STATEMENT

Pascal Bertram is a current employee of Brainlab AG, which manufactures the ExacTrac imaging system.

Daniel Saenz reports a research agreement with Brainlab ending in March 2020 for projects unrelated to the current manuscript.

DATA AVAILABILITY STATEMENT

Research data will be shared to the extent allowed under Institutional Review Board approval and institutional data-sharing agreements, upon request to the corresponding author.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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